



K. J. Somaiya School of Engineering

Batch: _____ Roll No.: _____
Experiment / assignment / tutorial No. 7
Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Active Filters

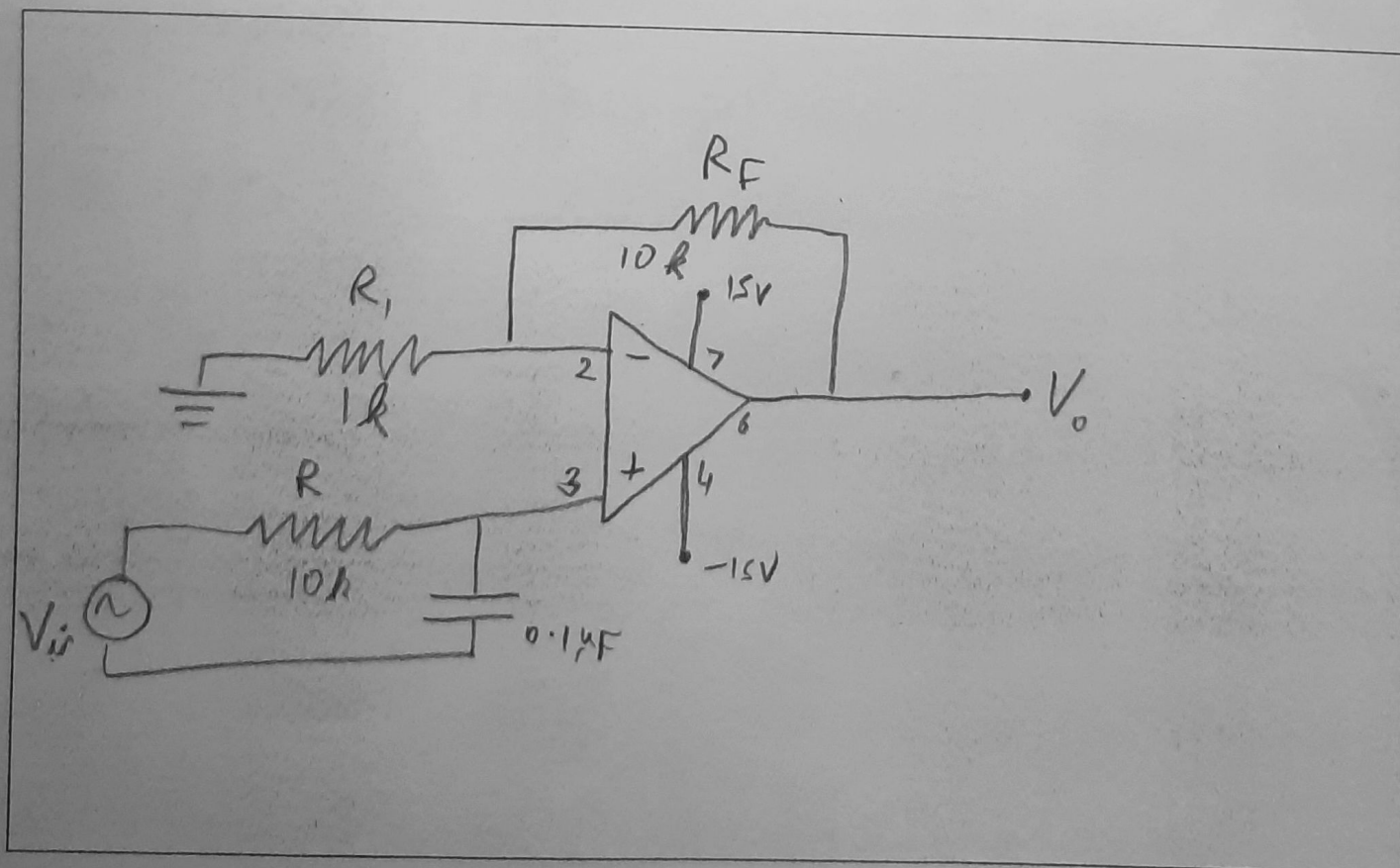
AIM: To implement and study active filters using OPAMP

OUTCOME: Students will be able to understand the basics of analog Integrated circuits.

PROCEDURE:

1. Assemble OPAMP filter circuit as shown in diagram.
2. Take sinusoidal input signal of fixed amplitude and apply it to the filter circuit.
3. Vary the frequency of the input signal and measure the output voltage.
4. Find frequency response of the circuits with reference to observation table I.
5. Plot frequency response, find cut-off frequency, and hence find the bandwidth.
6. Next, design a first order HPF for cut-off frequency of 10 kHz and pass band gain of 2.
7. Assemble the designed filter circuit and repeat steps 2 to 4. (Use Observation Table II.)

Circuit Diagram 1:



Observation Table I (circuit 1):

$$V_{CC} = 12 \text{ V}, V_{EE} = \text{V}$$

$$\text{Input Signal Voltage } (V_s) = 20 \text{ mV}$$

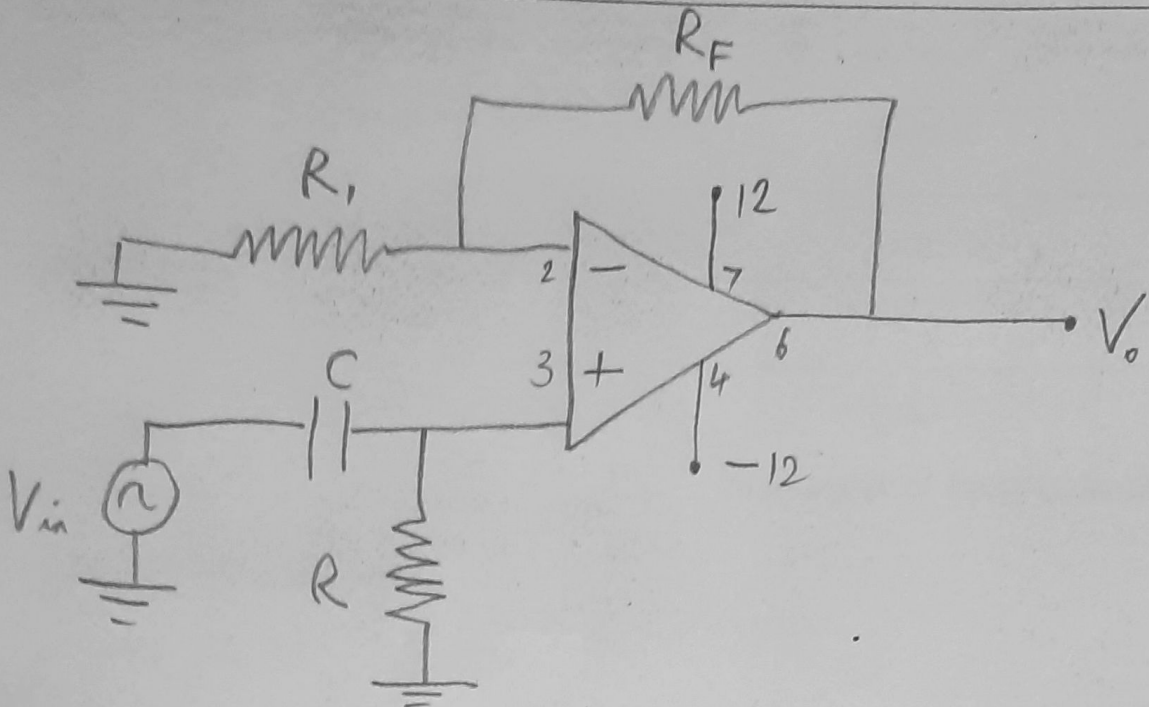
Sr. No.	Input Signal (V_s) frequency	Output Voltage (V_o)	Voltage Gain (V_o/V_s)	Gain in dB
1.	11	2.77	13.85	22.83
2.	12	2.85	14.25	23.07
3.	13	3.06 2.88	14.45	23.20
4.	14	3.06 3.06	15.30	23.69
5.	15	3.18 3.10	15.50	23.81
6.	16	3.22 3.18	15.90	24.03
7.	17	3.26 3.22	16.10	24.13
8.	18	3.30 3.26	16.30	24.24
9.	19	3.30	16.50	24.35
10.	20	3.30	16.50	24.35

Design Steps for HPF

- ① Given: Cutoff frequency, passband gain $A_v = ?$
- ② Use $f_c = \frac{1}{2\pi RC}$ choose $C = 0.01 \mu\text{F}$ and calculate $R = \frac{1}{2\pi f_c C}$
- ③ Set gain using op-amp non-inverting configuration: $A_v = 1 + \frac{R_f}{R_2}$
For gain 2, $R_f = R_1$
- ④ Verify design by calculating theoretical cutoff freq. and gain



Circuit Diagram 2:



Observation Table II (circuit 2):

$V_{CC} = 12\text{ V}$, $V_{EE} = \text{V}$ Input Signal Voltage (V_s) = 20 mV

Sr. No	Input Signal (V_s) frequency	Output Voltage (V_o)	Voltage Gain (V_o/V_s)	Gain in dB
1.	0.3	24	1.2	1.58
2.	0.5	32	1.6	4.08
3.	2	96	4.80	13.62
4.	3	140	7.00	16.90
5.	6	233	11.65	21.33
6.	8	270	13.50	22.61
7.	10	297	14.85	23.44
8.				

Observation Table III

	LPF		HPF	
	Practical	Theoretical	Practical	Theoretical
Cut off Frequency	160.20 Hz	159.15 Hz	10.04 kHz	10 kHz
Voltage Gain	10.58	11	2.32	2

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Theoretical Calculations:

For low pass filter

$$R_F = 10 \text{ k}\Omega \quad R = 10 \text{ k}\Omega \quad R_1 = 10 \text{ k}\Omega \quad C = 0.1 \mu\text{F}$$

$$\text{Cutoff freq. } f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 10 \times 0.1 \times 10^{-6}} = 159.15 \text{ Hz}$$

Gain for non inverting terminal

$$\text{Gain} = 1 + \left(\frac{R_F}{R_1}\right) = 1 + \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega} = 11$$

For high pass filter

Assume $C = 0.1 \mu\text{F}$, $f_c = 10 \text{ kHz}$

$$f_c = \frac{1}{2\pi RC} \quad \therefore R = \frac{1}{2\pi f_c C}$$

Conclusion: From the experiment, we can conclude that the low pass filter allows frequencies below f_c to pass through whereas high pass filter allows frequencies above f_c to pass through it.